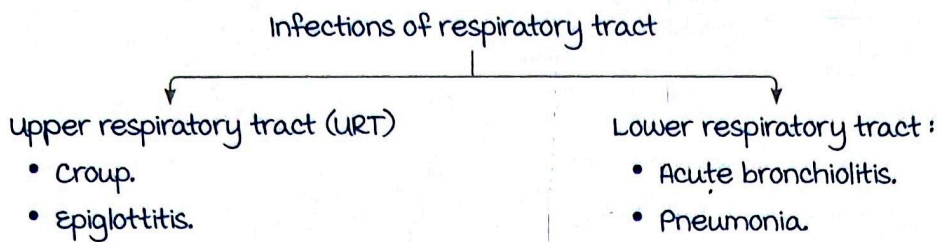


RESPIRATORY INFECTIONS

----- Active space -----



Laryngeal Infections

00:00:20

CROUP :

AKA acute laryngo-tracheo bronchitis.

Age : 6 months to 5 yrs Peak age : 2 yrs.

Etiology :

viral infection : **Parainfluenza virus** (m/c) — { Type 1 : m/c (Intermediate severity)
 Type 2 : mildest disease
 Type 3 : most severe disease

Clinical features :

Running nose, cough (URI symptoms)

↓ 2-3 d

- Severe cough : **Barking** / seal-like / brassy cough.
 - Inflammatory narrowing of :
 - Larynx : Stridor.
 - Trachea & bronchi : wheeze.
- } Child appears well despite symptoms (unless severe)

Severity classification :

	mild	moderate	Severe
Stridor	Only on exertion	Present at rest	Present at rest
Respiratory distress	Absent	Present	Severe
SpO ₂	Normal	Normal	Decreased (< 92 %)

X-ray findings :

Subglottic narrowed air column : **Steeple**/inverted-V sign.

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management :

- Primarily supportive (Self-limiting).
- Based on severity :
 - mild croup : Oral dexamethasone (Decrease inflammation).
 - moderate to severe :
 1. Dexamethasone.
 2. O_2 inhalation.
 3. Neb **Racemic** epinephrine : ↓ edema in pre-capillary arterioles.

↓
(1 : 1 D and L isomers)

ACUTE EPIGLOTTITIS :

Age : 3 - 6 yrs.

Etiology : Bacterial.

- **Hemophilus influenzae type B** (m/c in India).
- Streptococcus pyogenes.
- Streptococcus pneumoniae.

Clinical features :

Severe airway inflammation : medical emergency
(↑ risk of obstruction).

- Sick looking child.
- High grade fever.
- Stridor : D/t inflammation of supraglottic area.
- **3 D's :**

- **Dysphagia** - Drooling of saliva - **Dyspnea/respiratory distress**

Note : Low grade fever seen in viral infections.

Investigations :**1. X-ray findings :**

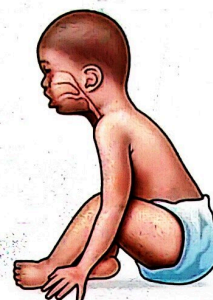
Thumb sign :
Soft tissue shadow of
swollen epiglottis.

2. Laryngoscopy findings :

- Inflamed cherry-red epiglottis.
- **Not done routinely :**

Stimulation by laryngoscope → Laryngospasm → Predisposes to Respiratory arrest

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Tripod posture : Leaning forward
opens up the airway

management:

----- Active space -----

1. Secure airway (Priority)

Intubation
 ↓ If not possible
 Tracheostomy

2. IV antibiotics : 3rd generation cephalosporin.

- IV ceftriaxone.
- IV cefotaxime.

Acute Bronchiolitis

00:14:21

Incidence :

- ↑ during winter.
- Age : < 2 yrs (Peaks 3 to 6 months)

Etiology : Respiratory Syncytial virus (RSV)

- Type A (more severe).
- Type B.

Note : RSV can cause URIs and LRIs.

Pathogenesis :

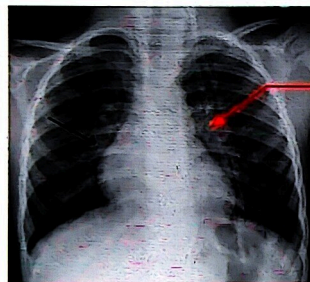
Inflammation of bronchioles → { Edema
 ↑ mucus production } → Airway obstruction → ↑↑ narrowing of airway during expiration → Air-trapping in lung fields

Clinical findings :

- D/t air trapping :
 - ↑ AP diameter.
 - Hyper-resonance on percussion.
- D/t narrowing bronchioles : wheeze.

Investigations :

1. X-ray findings :



a. Peribronchovascular cuffing d/t inflammatory exudates

b. Bilateral hyperinflation

2. Nasopharyngeal swab : PCR for RSV.

management :

usually self-limiting (3 to 7 days).

Supportive management :

- moist O_2 inhalation (Goal : $SpO_2 > 92\%$).
- Fluids (Oral or IV).

Prevention : For at-risk infants.

- Palivizumab : monoclonal antibody against RSV proteins.

Antivirals : Not routinely used.

- used in :

- Chronic lung disease.
 - Congenital heart defect (CHD).
 - Drug used : Inhaled Ribavirin (Nebulization).
- } At-risk infants

----- Active space -----

Pneumonia in Children

00:23:00

- Infection of lung tissue/parenchyma.
- Leading cause of under 5 mortality.

ETIOLOGY:

- | | | |
|---|---|---|
| Bacterial | > | Viral |
| <ul style="list-style-type: none"> • Streptococcus pneumonia (Overall m/c). • E. coli. • Group-B Streptococci. | } | <ul style="list-style-type: none"> • RSV (m/c) |
| (m/c in neonates) | | |

Note : Etiological diagnosis is not always feasible → X-ray findings more reliable.

CLINICAL FEATURES :

- Fever.
- Productive wet cough : may not be expectorated (D/t swallowing).
- Tachypnea.
- Crepitations on auscultation.

COMMUNITY MANAGEMENT GUIDELINES :

Guidelines given by : Integrated management of Neonatal & Childhood Illness (IMNCI). management of child with pneumonia in a peripheral health setting / at community level.

3 categories (Colour - coded)

	Features	management
No pneumonia	Fever, cough/cold	At home : Paracetamol (PCT) (Symptomatic)
Pneumonia	• Fever, cough/cold + fast breathing and/or + chest indrawing. • SpO₂ : Normal.	• At home : PCT + oral amoxicillin x 5 d. - If wheeze + : Salbutamol x 5d. • If cough > 2 wk : Evaluate for tuberculosis.
Severe pneumonia AKA very severe disease	• SpO₂ < 90 % OR, • ≥ 1 danger signs : - Lethargy. - Not feeding. - Convulsions. - Cyanosis.	urgent referral to tertiary care hospital + 1 st dose : IV antibiotic (Ampicillin + gentamycin)

Fast-breathing criteria :

----- Active space -----

Age	RR
< 2 months	> 60/min
2 - 12 months	> 50/min
> 12 months	> 40/min

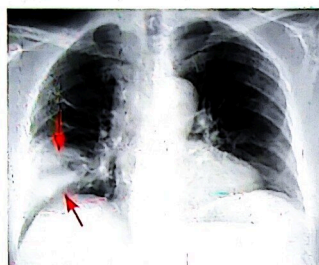
PNEUMOCOCCAL PNEUMONIA :

mode of transmission : Droplet.

Season of ↑ incidence : Winter

Features :

1. Lobar consolidation : On chest x-ray.
2. Clinical findings :
 - ↓ air entry.
 - Crepitations.
 - Sputum (if expectorated) : Rusty.
 - Bronchial sounds.
 - Dull /impaired note on percussion.



Lobar consolidation

Treatment :

- Penicillin G : 50,000 IU/ kg/ day IV for 5-7 days.
 ↓ If not available
- 3rd generation cephalosporins :
 - IV ceftriaxone.
 - IV cefotaxime.

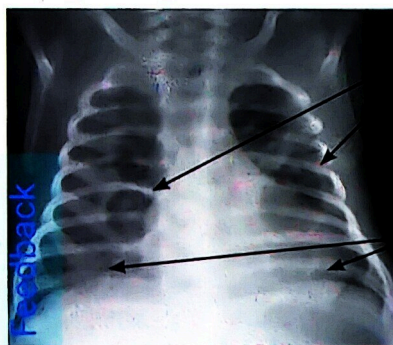
STAPHYLOCOCCAL PNEUMONIA :

Bacteria : Staphylococcus aureus.

Staphylococcal etiology seen in :

- Spread from different focus of infection : Pyoderma (Skin).
- Secondary to viral infection (measles, influenza etc).
- Cystic fibrosis (Recurrent pneumonia).

X-ray findings :



Pneumatocele : Air-filled cystic spaces Rupture → Pneumothorax (Pathognomonic)

B/L patchy consolidation (with pus)

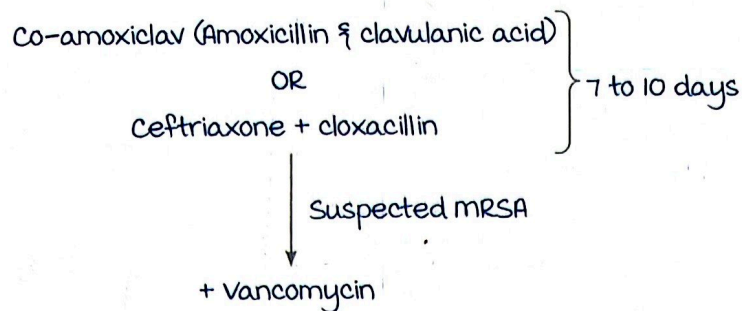
Empyema (If seen in child < 2 yr → m/c staphylococcal etiology)

Purulent pericarditis



----- Active space -----

Treatment : Antibiotics.

**PRIMARY ATYPICAL PNEUMONIA :**Organisms : m/c *Mycoplasma*.

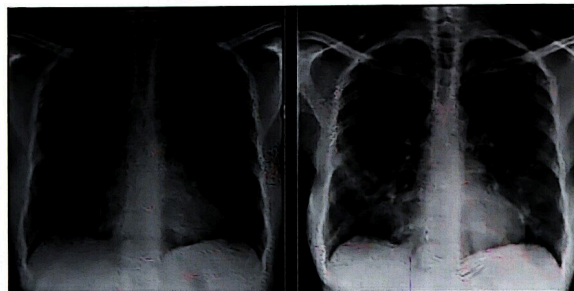
Age : > 5 yr (Rarely seen < 5 yr).

Features :

- Low grade fever.
- Cough.
- Scattered crepitations.
- Hemolysis may be seen.

Investigations :

1. X-ray findings : more significant than clinical findings.
2. Cold agglutination test : Positive.
 - IgM against RBC → Hemolysis.



Interstitial pneumonia

Treatment :

macrolide antibiotics (Oral).

- Azithromycin OR,
 - Erythromycin.
- } 7 to 10 days